

Mohamed AlKharraz

+973 3330 0673

alkharraz.mohd@gmail.com

alkharraz.netlify.app

linkedin.com/in/malkharraz



PROFESSIONAL PROFILE

Digital Solutions Engineer in oil and gas, bridging field automation with data-driven optimization. Over 5 years of experience across Bahrain and Saudi Arabia, building high-impact solutions spanning production optimization, surveillance, and real-time analytics. Technical expertise spans artificial lift systems including rod lift and gas lift, data science, AI/ML, SCADA, OT cybersecurity, and IoT Controllers; PMP-certified. Proven track record of leading, implementing, and advising across oil and gas projects that deliver measurable operational and production impact.

EXPERIENCE



Weatherford

Digital Solutions Engineer | October 2022 - Present

- Deployment support engineer at Bapco Upstream, providing technical support, resolving issues, and maintaining systems to ensure seamless operations.
- Appointed as Ethics Ambassador, reinforcing Code of Business Conduct, ethics, and compliance.
- Deployed an Industry 4.0 well optimization platform and SCADA across 3,000+ wells in the Bahrain Oil Field.
- Delivered an intelligent well test management platform across 48 test facilities with zero operational downtime, improving well test quality, scheduling visibility, and production surveillance.
- Broad knowledge of Artificial Lift Systems - Rod Lift, Gas Lift, Natural Flow, PCP, ESP, Plunger Lift.
- Engineering in-house automation solutions, reducing reliance on external vendors to cut costs.
- Working with customers to develop new technology to improve system productivity.
- Facilitating alignment between product managers, designers, QA teams, and stakeholders to ensure product developments meet business requirements.
- Experience in designing, developing, architecting, implementing, and supporting end-to-end systems.



Bapco Upstream

Service Engineer | October 2022 - Present

- Providing daily operational support for ForeSite, CygNet, PI Systems, LOWIS, including proactive monitoring, troubleshooting, user assistance, and implementing system upgrades and enhancements.
- Configuring, maintaining and troubleshooting hardware including: IoT controllers, WellPilot, LufkinSAM, ForeSite Edge.
- Develop, analyze, and manage business and operational reports using BI dashboards and scripting.
- Manage data in OT/IT environments ensuring unidirectional information exchange through data diode.
- Collaborating with cross-disciplinary teams to successfully integrate various reports, synchronize information across integrated systems, improving data accuracy and integrity.
- Ensured data accuracy through rigorous quality checks and established governance frameworks to standardize data across the organization.



Tatweer Petroleum

AI Engineer & Data Science Intern | September 2021 - August 2022

- Developed machine learning models to forecast oil production using sensory data.
- Created business intelligence dashboards, visualizations, and reports to provide actionable insight.
- Increased operations productivity by building workflows to monitor equipment downtime.



Atlas Copco

Intern | June 2021 - August 2021

- Worked on compressed air systems design, gas generation, piping, AIRnet pipes, Atlas Copco Optimizers, desiccant dryers, and ventilation design.

EDUCATION



University of Bahrain

B.Sc. in Mechanical Engineering | September 2017 - August 2021

- Graduated with excellence (GPA: 3.5 / 4.0).

CERTIFICATES



Project Management Professional (PMP) | PMI | March 2026



Professional Excellence Program | WISC | November 2025



Well Control | IADC | October 2023



The Complete SQL Bootcamp: Go from Zero to Hero | Udemy | August 2023



Alarm Management (ISA-18.2 / IEC 62682) | Exida | July 2023



AWS Certified Machine Learning Specialty | AWS | April 2022



IBM AI Engineering Professional Certificate | IBM | January 2022



Machine Learning | Stanford Online | July 2021



Engineering Project Management Specialization | Rice University | August 2020



The Finite Element Method for Problems in Physics | Michigan Univ. | March 2020



Basics of Arduino | YC2030 | 2014

PROJECTS

Advanced Inferred Production | 2026

- Developed a reliable virtual production rate system for rod lift wells, enabling real-time production estimation.
- Redefined well test strategy for rod lift wells by reducing reliance on third-party well testers.

Intelligent Well Test Management Platform for Bapco Upstream | 2026

- Led end-to-end delivery as project manager and technical implementer, overseeing deployment across 48 test facilities with zero operational downtime.
- Improved well test quality and production surveillance; built a smart well prioritization system, faulty-valve identifier, and a schedule tracker with forward visibility and root-cause documentation of missed tests.

Industry 4.0 Digital Transformation of Bahrain Oil Field Production Infrastructure | 2023

- Deployed a well production optimization platform & SCADA system across 3,000+ wells.

- Integrated with business databases and enterprise systems; introduced automated workflows, operational dashboards, and production reports to deliver actionable field intelligence across the organization
- IT/OT network segmentation with unidirectional data transfer to achieve full OT cybersecurity compliance.

Gas Compressors Availability Reporting & Workflow Optimization | 2022

- Led end-to-end development, from data entry GUI to creating Power BI dashboards and summary reports.
- Leveraged data analysis to swiftly identify and resolve availability issues.

PROFESSIONAL SKILLS

- **Programming:** Python | C | C++ | SQL | VBA | Linux | PowerShell | HTML | Workflow Automation.
- **Software:** OSI PI | ForeSite | CygNet | LOWIS | PAS | AWS | Power BI | SQL SSRS | Spotfire | Grafana.
- **Hardware:** WellPilot | IOT Edge Devices | RTUs | IOT Sensors | LufkinSAM | PLC | DNOC | Gas lift controllers.
- **Analytical:** Data Analysis Modelling | Data Integration | Computational fluid dynamics | Optimization.
- **Language:** English (Fluent) | Arabic (Native) | French (Intermediate) | Spanish (Intermediate).

AWARDS

- Kanoo Engineering Excellence Award 2021
- Weatherford Ethics Ambassador 2024 - 2026

Memberships

- Society of Petroleum Engineers
- International Society of Automation (ISA)

PUBLICATIONS

- "Autonomous Control for Gas Lift Wells on Edge." GIITS, 2023.
- "Data Driven Autonomous Control Optimization." GOTECH, 2026.

REFERENCES

- **Ali Sharaf** | Associate Director - Data Science at Tamkeen. E-mail: Ahsharaf@tamkeen.bh
- **Dr. Omar Al-Abbasi** | Head of Mechanical Eng Dept - University of Bahrain. E-mail: omabbasi@uob.edu.bh